



DELHI PUBLIC SCHOOL NEWTOWN  
SESSION 2024-25  
PRE BOARD EXAMINATION

CLASS: X  
SUBJECT: PHYSICS [SET A]

FULL MARKS: 80  
TIME: 2 HOURS

*Answers to this Paper must be written on the paper provided separately.*

*You will not be allowed to write during the first 15 minutes.*

*This time is to be spent in reading the question paper.*

*The time given at the head of this Paper is the time allowed for writing the answers.*

*Section A is compulsory. Attempt any four questions from Section B.*

*The intended marks for questions or parts of questions are given in brackets [ ].*

*This paper consists of eight printed pages*

**SECTION A**

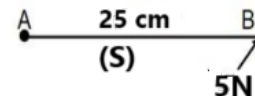
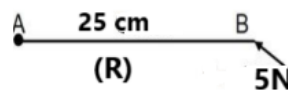
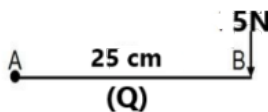
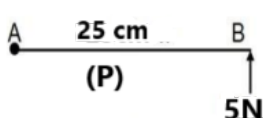
**Question 1**

[15 × 1 = 15]

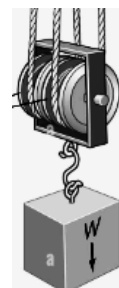
Choose the correct answers to the questions from the given options:

(Do not copy the question, write the correct answers only.)

- (i) The diagrams below show 5N force is acting on a rod AB pivoted at A in different directions. Identify the correct statement.

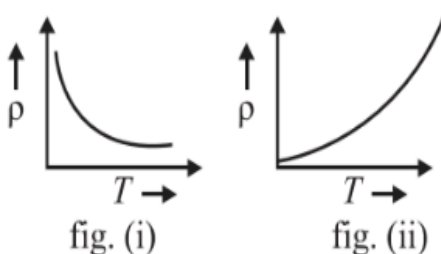


- (a) the moment of force in R is negative.  
(b) the magnitude of the moment of force is maximum in R.  
(c) P and Q have opposite moments.  
(d) the magnitude of the moment of force is maximum in P and Q.
- (ii) A solid cone A, a hollow cone B, and a solid cylinder C, all have the same height of 12 cm. The positions of their centers of gravity from the base are calculated. Which of the following shows the correct sequence of the positions of the centers of gravity from the lowest to the highest?  
(a)  $A < B < C$       (b)  $B < A < C$       (c)  $C < A < B$       (d)  $A < C < B$
- (iii) The energy change that takes place in a thermocouple:  
(a) electrical to heat      (b) mechanical to heat  
(c) heat to mechanical      (d) heat to electrical
- (iv) The adjacent diagram shows the movable block of a block and tackle system. The direction of effort is not specified. What could be the probable mechanical advantage of the system?



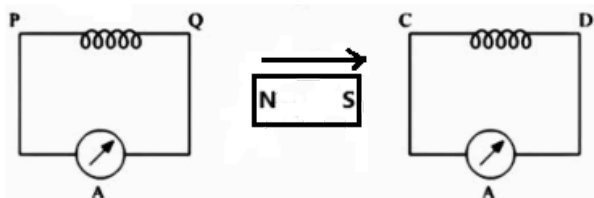
- (a) 4      (b) 5      (c) 3      (d) both (a) and (b)

- (v) A star appears twinkling in the sky. This phenomenon is due to :  
 (a) reflection of light (b) refraction of light  
 (c) total Internal Reflection (d) dispersion of light
- (vi) The critical angle for water-air is  $49^\circ$  for green light. The critical angle for red light will be:  
 (a) less than  $49^\circ$  (b) more than  $49^\circ$   
 (c) equal to  $49^\circ$  (d) insufficient data
- (vii) The focal length of a concave lens is 20 cm. What will be its maximum image distance?  
 (a) 10 cm  
 (b) 20 cm  
 (c) anywhere between 10 cm and 20 cm.  
 (d) cannot be determined from the given information
- (viii) Assertion : The focal length of a lens does not change when red light is replaced by blue light.  
 Reason: The focal length of a lens depends on the refractive index of material of lens.  
 (a) both A and R are true and R is the correct explanation of the A.  
 (b) both A and R are true and R is not the correct explanation of the A.  
 (c) A is false but R is true.  
 (d) A is true but R is false.
- (ix) String X is thicker than String Y. If both are under the same tension, which of the following is true for them to produce the same pitch?  
 (a) string X must be longer than String Y.  
 (b) string X must be shorter than String Y.  
 (c) both strings must have the same length.  
 (d) the length of the string does not matter.
- (x) The temperature (T) dependance of resistivity of materials A and B is represented by fig (i) and fig (ii) respectively. Identify material A and material B.



- (a) A- copper, B- germanium (b) A- germanium, B-copper  
 (c) A-nichrome, B-germanium (d) A-copper, B-nichrome
- (xi) In electrical wiring, specific colors are used to indicate the function of different wires. Which of the following statements correctly describe the standard color coding for electrical wires?  
 (a) the yellow wire is used as the neutral wire, while the blue wire is the live wire.  
 (b) the green wire is used for grounding or earthing, while the blue wire is used as the neutral wire.  
 (c) the black wire is used as the live wire, while the red wire is used for neutral.  
 (d) the brown wire indicates the ground connection, and the red wire is used for the live connection.

- (xii) A bar magnet is moved in the direction indicated by the arrow between two coils PQ and CD. Predict the directions of induced current in each coil.



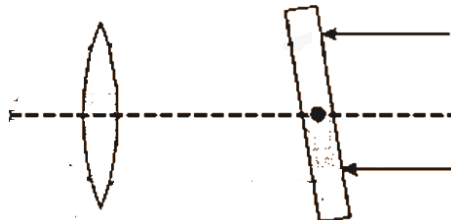
- (a) anticlockwise in PQ and clockwise in CD  
 (b) clockwise in PQ and clockwise in CD  
 (c) anticlockwise in PQ and anticlockwise CD  
 (d) clockwise in PQ and anticlockwise in CD
- (xiii) Same amount of heat is added to two substances A and B of the same mass having specific heat capacities  $s_1$  and  $s_2$  such that  $s_1 > s_2$ . Select the correct statement.  
 (a) the change in temperature of A is more than the change in temperature B.  
 (b) the change in temperature of A is less than the change in temperature B.  
 (c) both will have the same change in temperature.  
 (d) temperature changes are not related to the specific heat capacities.
- (xiv) You are given two identical blocks of different substances, one made of ice and the other made of wax, both at their respective melting points. If the same amount of pressure is applied to both, which option is correct for the melting points of ice and wax?  
 (a) decreases, increases  
 (b) increases, decreases  
 (c) both decreases  
 (d) both increases
- (xv)  $^{23}\text{Na}_{11}$  and  $^{23}\text{Mg}_{12}$  are the examples of;  
 (a) isotones (b) isotopes (c) isobars (d) both isotones and isobars

## Question 2

- (i) Complete the following by choosing the correct answers from the bracket: [6]
- (a) The electricity meter installed in our home measures electric energy in the units of \_\_\_\_\_ [kW/kWh/J]  
 (b) A movable pulley is used as a \_\_\_\_\_ [force/ energy/speed] multiplier.  
 (c) A total reflecting equilateral prism can be used to deviate a ray of light through \_\_\_\_\_ [30°/60°/90°]  
 (d) Two copper wires having radii 2 mm and 3 mm, will have the ratio of resistance \_\_\_\_\_ [4:9, 2:3, 9:4].  
 (e) The image is formed \_\_\_\_\_ [at  $F_2$  /at  $2F_2$  /beyond  $2F_2$ ] in a terrestrial telescope, but in a slide projector the image is formed \_\_\_\_\_ [at infinity/at  $2F_2$  /beyond  $2F_2$ ].
- (ii) Draw a graph showing the variation of the resistance of a wire with the following, when other variables remain constant : (a) radius (b) length [2]
- (iii) (a) Name the radiations used for studying the atomic arrangement in crystals. [2]  
 (b) Mention one of its properties.

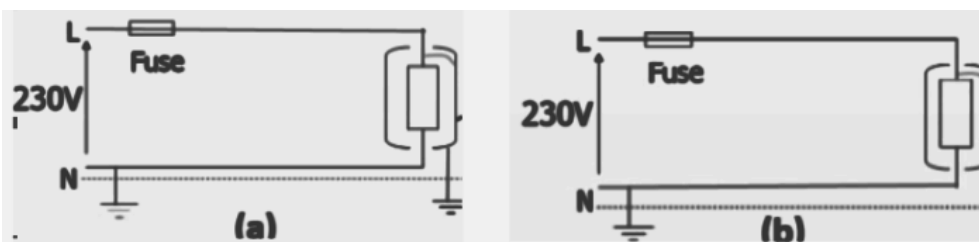
### Question 3

- (i) (a) A glass slab is placed in the path of a parallel light beam before it is passing through the lens, as shown in the figure. Where will the rays meet after passing through the lens. [2]



- (b) The power of a lens is +2.0 D. Identify the lens.

- (ii) In the following diagram, two people are touching similar appliances. Identify in which diagrams- (a) or (b) the person is getting a shock? Justify your answer. [2]



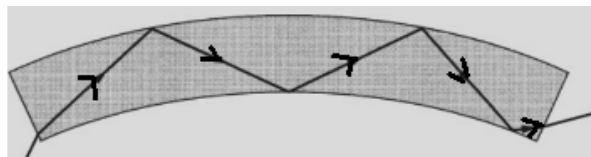
- (iii) We always connect a charger to mains to charge our mobile phones. A transformer is used for the purpose. [2]  
 (a) Identify the type of transformer used for the above purpose.  
 (b) State its turns ratio.
- (iv) A solid of mass 80 g at  $100^{\circ}\text{C}$  is placed in 55 g of water at  $35^{\circ}\text{C}$ . The final temperature is  $45^{\circ}\text{C}$ . Calculate the heat capacity of the solid. [sp. heat capacity of water is  $4.2 \text{ Jg}^{-1}\text{K}^{-1}$ ] [2]
- (v) (a) What is the function of the split ring commutator in a DC motor? [2]  
 (b) Name the SI unit of the strength of the magnetic field.
- (vi) (a) What is the origin of  $\beta$ -particles? [2]  
 (b) Show its emission by means of an equation.
- (vii) Tara, after completing her exams, went to her grandmother's house along with her younger brother Rahul. One day, Rahul came home after playing and switched on the fan and air conditioner. At the same time, his grandmother was preparing a milkshake for him using a mixer. Suddenly, they heard a big sound and the electricity in the house got shut down. Tara called an electrician. He said that the main reason for the supply to shut down was overloading.  
 (a) What is overloading?  
 (b) Are overloading and short circuits the same thing?  
 (c) When does short circuit happen? [3]

**SECTION B**  
(Attempt any four questions)

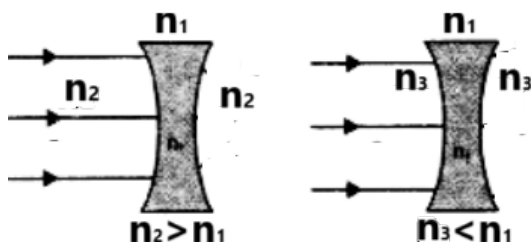
**Question 4**

[3+3+4]

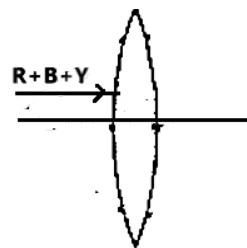
- (i) Following is a diagram of a fiber-optic device used in medical applications like endoscopy.



- (a) Name the phenomenon that takes place here.  
 (b) In the following diagram, a double concave lens with refractive index  $n_1$  is kept inside two liquids of refractive indices  $n_2$  and  $n_3$ . How would a parallel incident beam of light refract when it comes out of the lens in each of the cases below?  
 (A)  $n_2 > n_1$                       (B)  $n_3 < n_1$



- (ii) The lateral shift of a glass slab for refractive index  $\mu_1$  is 1.5 cm. When the same thing was checked by another glass slab of refractive index  $\mu_2$ , lateral shift changes to 'y'.  
 (a) State whether  $y < 1.5$  cm or  $y > 1.5$  cm if  $\mu_1 < \mu_2$   
 (b) State two differences between total internal reflection and reflection from a plane mirror.
- (iii) (a) A beam of light consisting of red, blue and yellow colours is incident on a convex lens, as shown in the figure below. State, whether the rays meet at a single point or at different points on the principal axis after emergence.  
 (b) Name the invisible radiations that are obtained with rock salt prism.  
 (c) State one of its uses.  
 (d) Name one radiation having the wavelength longer than the wavelength of these radiations.



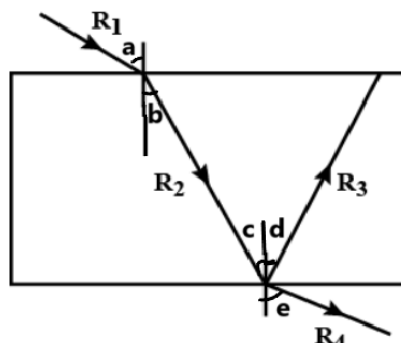
**Question 5**

[3+3+4]

- (i) An object is placed at a distance 18 cm in front of a convex lens of focal length 12 cm.  
 (a) What is the nature of the image so formed?  
 (b) Calculate image distance and its magnification.
- (ii) (a) Name the process behind the formation of the red colour of the Sun at sunrise. State the condition on which this process occurs.  
 (b) Provide an example where the condition mentioned above is not valid.

- (iii) A ray  $R_1$  is incident on the plane surface of the glass slab (kept in air) of refractive index  $\sqrt{2}$  at an angle of incidence 'a'. The refracted ray  $R_2$  undergoes partial reflection and refraction at the other surface. Complete the following table with the reference to the question and the diagram. One is done for you.

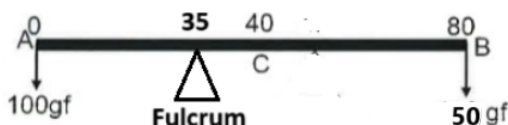
| sr. no. | quantity                                   | value      |
|---------|--------------------------------------------|------------|
| 1.      | a                                          |            |
| 2.      | d                                          | $30^\circ$ |
| 3.      | b                                          |            |
| 4.      | e                                          |            |
| 5.      | angle of deviation between $R_2$ and $R_4$ |            |



#### Question 6

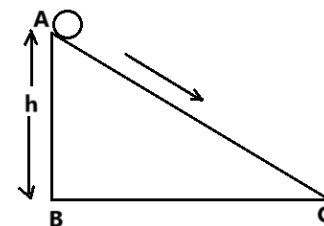
[3+3+4]

- (i) A metal rod AB of length 80 cm is balanced at 35 cm from the end A with weights suspended from the two ends as shown below.



- (a) Find the weight of the rod.  
 (b) State the principle that you have used here.
- (ii) For each of the following situations, state whether the work done by gravity is positive, negative or zero.
- satellite orbiting the earth.
  - throwing a ball upward.
  - a child sliding down a slide.

- (iii) A ball of mass 2 kg slides down the inclined plane as shown in the figure.



- The maximum velocity acquired by the ball is 6 m/s. Find the height 'h'. [ $g=10 \text{ N/kg}$ ]
- What is the total energy at the mid point of AC?
- Draw an energy versus distance graph as the ball falls down from A to C and mark kinetic energy and potential energy in the graph.

#### Question 7

[3+3+4]

- (i) A block and tackle pulley system containing 4 pulleys has efficiency 80%. A load of 320 kgf is lifted by the pulley system.

- (a) Find the effort needed to overcome the load.

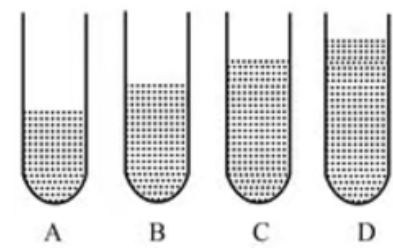
(b) You can reduce friction to increase the efficiency for this design. What else can you do to get greater efficiency for this design?

- (ii) A person stands near a large wall and shouts, hearing the echo after 4 seconds. If the speed of the sound in the air is 343 m/s.

(a) Calculate the initial distance between the person and the wall.

(b) The person then moves farther away from the wall, and when they shout again the echo takes 1.5 seconds longer than before. Calculate how much further the person has moved from the wall.

- (iii) John was doing an experiment in the laboratory with four test tubes A, B, C and D, each with a height of 20 cm. He filled each tube with water at different levels, A - 12 cm, B - 14 cm, C - 16 cm, D - 18 cm respectively. He then took a vibrating tuning fork and placed it near the mouth of test tube D and heard a loud sound. He tried the same thing with other tubes, however he didn't hear any loud sound. Answer the following:



(a) Name the phenomenon that happens with tube D.

(b) Why was the loud sound heard with test tube D?

(c) Name the phenomenon that occurred with other tubes.

(d) What needs to be done if he would want to hear a loud sound from all the tubes?

### Question 8

[3+3+4]

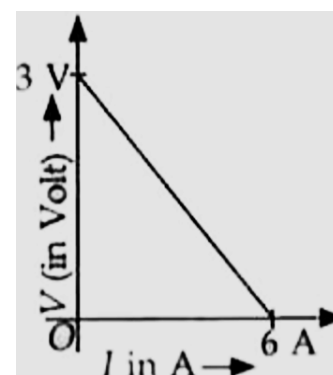
- (i) The following graph shows the variation of terminal potential difference  $V$ , across a cell to a resistor, versus current  $I$ .

Calculate:

(a) emf

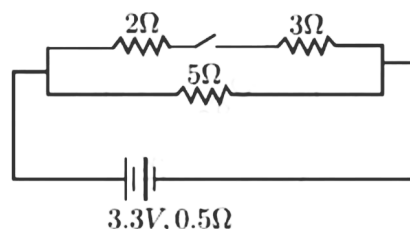
(b) internal resistance

(c) terminal voltage when current is 1.5 A.



- (ii) (a) Distinguish between radioactive decay and nuclear fission.  
(b) Why is U-235 preferred over U-238 as a fissionable material?

- (iii) The diagram alongside shows a circuit with the key K open.



Calculate:

(a) the current drawn from the cell when the key K is open.

(b) the equivalent resistance when the key K is closed.

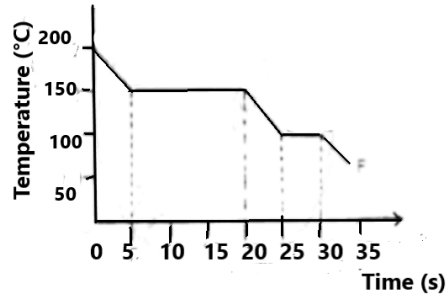
(c) the current drawn through 5 ohm resistance when the circuit is closed.

### Question 9

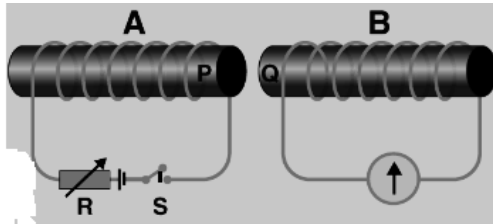
[3+3+4]

- (i) What mass of ice at  $0^{\circ}\text{C}$  needs to be added to 3 g of water at  $80^{\circ}\text{C}$  to cool it down to  $30^{\circ}\text{C}$ ? Given: specific heat capacity of water =  $4.2 \text{ J/g}^{\circ}\text{C}$ . Specific latent heat of ice =  $336 \text{ J/g}$ .

- (ii) The diagram below shows a cooling curve for a liquid of mass 10 kg. Heat is transferred at  $10^6 \text{ J/s}$ .



- (a) Mention the range of temperature in which it is in liquid state.  
(b) Calculate the specific latent heat of condensation of the liquid.  
(c) Why is steam at  $100^\circ\text{C}$  more painful, causing burns than boiling water at the same temperature?
- (iii) The diagram below shows two solenoids A and B. A is connected with a rheostat, cell, and a switch. B is connected with a galvanometer.



- (a) When will you observe a deflection in B?  
(b) Mention one factor to change the amount of deflection.  
(c) Name the law that you have used to find the direction of deflection of the galvanometer.  
(d) Name one device that works with this set up.